

# Split and insert

Revise 360 lesson plan — 60 minutes

|                 |                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| Unit            | OCR J277 2.1 Algorithms — Paper 2, Computational thinking, algorithms and programming                             |
| Lesson          | Split and insert (lesson 9 of the unit)                                                                           |
| Lesson ID       | a1-l09                                                                                                            |
| Experience ID   | a1-l09                                                                                                            |
| Plan ID         | PLAN-AL-L09                                                                                                       |
| Content version | 0fa4362382d9                                                                                                      |
| Website         | <a href="https://revise360.co.uk/experience.html?id=a1-l09">https://revise360.co.uk/experience.html?id=a1-l09</a> |

## Big question, outcomes and success criteria

**Big question** — How do merge sort and insertion sort work, and how do they compare with a bubble sort?

### Prerequisites

The lesson before this one is a1-l08. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: a1-l01, a1-l02, a1-l03, a1-l04, a1-l05, a1-l06, a1-l07.

### Assessable outcomes

1. **a1-l09-o1** — understand the merge sort algorithm
2. **a1-l09-o2** — understand the insertion sort algorithm
3. **a1-l09-o3** — be able to compare the three sorting algorithms

## Vocabulary, resources and preparation

### Key vocabulary — Merge sort, Insertion sort

### Resources

|                   |                                                 |
|-------------------|-------------------------------------------------|
| SLIDES            | presentations/AL_L09_SplitAndInsert_Lesson.pptx |
| WORKSHEET         | worksheets/AL_L09_SplitAndInsert_Worksheet.docx |
| ONLINE EXPERIENCE | experience.html?id=a1-l09                       |

### Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

### How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.

3. Choose the topic, then this lesson's card.

**Worth knowing before you promise anything.** In this build APP\_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

## Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

| MIN   | PHASE                                     | TEACHER                                                                                      | PUPILS                                                                          | EVIDENCE TO CHECK                                                                      |
|-------|-------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 0-5   | <b>Retrieve</b>                           | Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites. | Answer the starter on the worksheet.                                            | The starter box on AL_L09_SplitAndInsert_Worksheet.docx is filled in.                  |
| 5-8   | <b>Review the starter</b>                 | Take answers; correct the one most got wrong.                                                | Correct their own starter.                                                      | A correction written, not just a tick.                                                 |
| 8-18  | <b>Explain and model</b>                  | The new idea in two or three steps, with the example worked in front of the class.           | Listen and answer questions about the example.                                  | They can say why, not only what.                                                       |
| 18-21 | <b>Set up and settle</b>                  | Sign-in, launch, and what to write before anyone taps an answer.                             | Sign in, open the lesson, find Merge sort: splitting.                           | Everyone is in and on station 1.                                                       |
| 21-43 | <b>Online: work the stations in order</b> | Pupils read each wall, write the key fact and the challenge, then answer at the badge.       | Read each wall, write the key fact and the challenge, then answer at the badge. | The worksheet has writing on it for each of the 6 stations before the badge is tapped. |
| 43-48 | <b>Check understanding</b>                | The hinge question, hands down.                                                              | Answer the hinge question, hands down.                                          | The split across the options.                                                          |
| 48-56 | <b>Apply</b>                              | Exam practice on the worksheet, in silence.                                                  | Exam practice, in silence.                                                      | worksheet, Exam practice a) to c)                                                      |
| 56-60 | <b>Exit and set up the next lesson</b>    | The exit question and the retrieval link to next lesson.                                     | Write the exit answer.                                                          | One sentence that answers the big question.                                            |

## The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

### 1. Merge sort: splitting

- Split the list in half, then split each half, and keep going.
- Stop when every piece holds a single item, which is sorted by definition.
- This is the divide part of divide and conquer.

**Challenge on this wall:** Show how 8, 3, 5, 1 is split down to single items.

### 2. Merge sort: merging

- Take two sorted pieces and combine them into one sorted piece.
- Compare the first item of each, take the smaller, and repeat.

- Keep merging pairs until one sorted list remains.

**Challenge on this wall:** Merge 3, 8 with 1, 5 and show each comparison.

### 3. Merge sort: the whole thing

- Split all the way down, then merge all the way back up.
- It is much faster than a bubble sort on large lists.
- It needs extra memory to hold the pieces, so it does not sort in place.

**Challenge on this wall:** Describe merge sort in two sentences: one for splitting, one for merging.

### 4. Insertion sort

- Treat the first item as a sorted section of one.
- Take the next item and move it left until it is in the right place within that section.
- Repeat until every item has been inserted.

**Challenge on this wall:** Run an insertion sort on 4, 2, 7, 1 and show each step.

### 5. Comparing the three sorts

- Bubble: simplest, slowest, sorts in place.
- Insertion: simple, faster than bubble, sorts in place, very fast on nearly sorted data.
- Merge: most complex, much faster on large lists, needs extra memory.

**Challenge on this wall:** Make a table comparing the three sorts on speed, memory and complexity.

### 6. Answering sorting questions

- Name the algorithm and describe its distinctive action: swap neighbours, insert into a sorted section, or split and merge.
- If asked to show the steps, show the list after each pass or insertion.
- Comparisons usually mean 'when does it stop' and 'how much memory'.

**Challenge on this wall:** Plan a six-mark answer comparing bubble sort and merge sort.

## Retrieval and the check for understanding

**Starter** — Deal yourself seven playing cards. Sort them into order and write down exactly what you did. Which algorithm was it closest to?

Answer space: 4 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

### Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (a1 - 109 - s3 - c2), so the wording a pupil sees matches the wording you say.

**Why is merge sort much faster than bubble sort on large lists?**

**Correct:** The work grows far more slowly as the list gets longer

### What a pupil might choose instead

- It makes no comparisons
- It uses no memory
- It sorts in one pass

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

**The course's own response to a wrong answer here:** For a million items the difference is minutes against days.

**One response, not four.** one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

## Guided to independent practice

| STAGE       | ACTIVITIES | WHAT IT IS                                                                                                                |
|-------------|------------|---------------------------------------------------------------------------------------------------------------------------|
| Recognise 7 |            | The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.                   |
| Construct 7 |            | Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer. |

### Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

### Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

## Delivering the online experience

### Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

### The stations, and what to record at each

| #     | STATION                      | ACTIVITIES | RECORD ON PAPER                           |
|-------|------------------------------|------------|-------------------------------------------|
| 1     | Merge sort: splitting        | 2          | Merge sort splits by...                   |
| 2     | Merge sort: merging          | 2          | Merge sort merges by...                   |
| 3     | Merge sort: the whole thing  | 2          | A merge sort works by...                  |
| 4     | Insertion sort               | 2          | An insertion sort works by...             |
| 5     | Comparing the three sorts    | 2          | The three sorts compare as...             |
| 6     | Answering sorting questions  | 2          | To answer a sorting question, I should... |
| final | Final challenge: sort it out | 2          | the answer to the station's challenge     |

### Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

### Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

## Assessment, evidence and unfinished work

| ID        | QUESTION                                                               | MARKS | WHERE                                                  |
|-----------|------------------------------------------------------------------------|-------|--------------------------------------------------------|
| a1-109-x1 | Describe how an insertion sort works.                                  | 3     | worksheet, Exam practice a); deck, Exam practice slide |
| a1-109-x2 | Show the stages of a merge sort on the list 6, 2, 9, 4.                | 4     | worksheet, Exam practice b); deck, Exam practice slide |
| a1-109-x3 | Compare a bubble sort with a merge sort, giving one advantage of each. | 4     | worksheet, Exam practice c); deck, Exam practice slide |

### Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/a1-109.json`, which is not published with the website.

### Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **Why does splitting stop at single items?** — Because the computer runs out of memory; Because two items cannot be compared
- **When merging two sorted lists, which item is taken next?** — The first item of the left list; The largest item overall
- **What is the main disadvantage of a merge sort?** — It only works on sorted lists; It is slower than a bubble sort
- **Why is merge sort much faster than bubble sort on large lists?** — It makes no comparisons; It uses no memory
- **Where does an insertion sort put each new item?** — At the end of the list; At the start of the list

### Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

## Exit, next lesson and homework

### Exit check

One of these, written, not a show of hands:

- Merge sort: splitting and merging
- Insertion sort
- Comparing the three sorts

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

### Retrieval link to next lesson

Next is a1-110, Fault finder. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

### Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

## Shorter lessons, longer lessons and no internet

### **A 40 to 45 minute lesson**

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

### **If the internet or the devices are unavailable**

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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